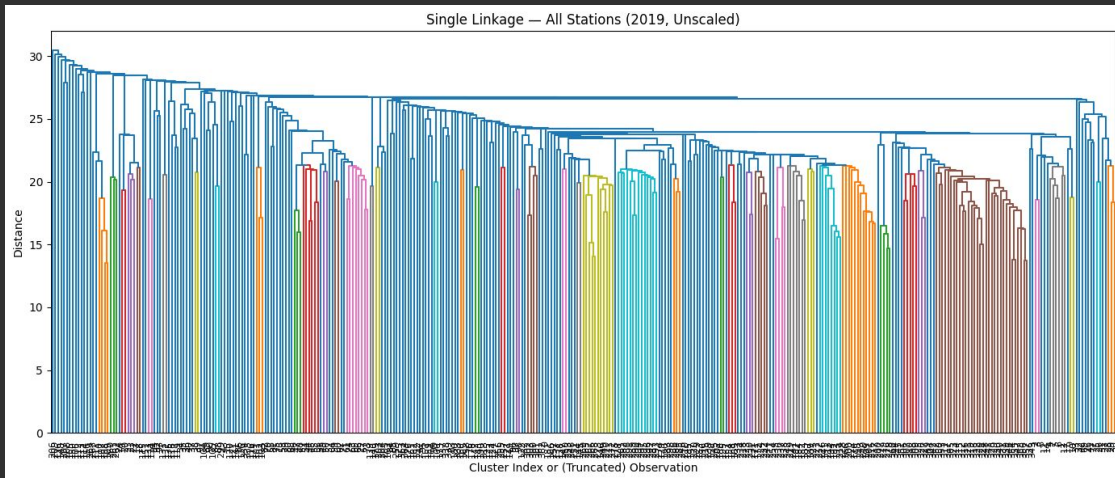
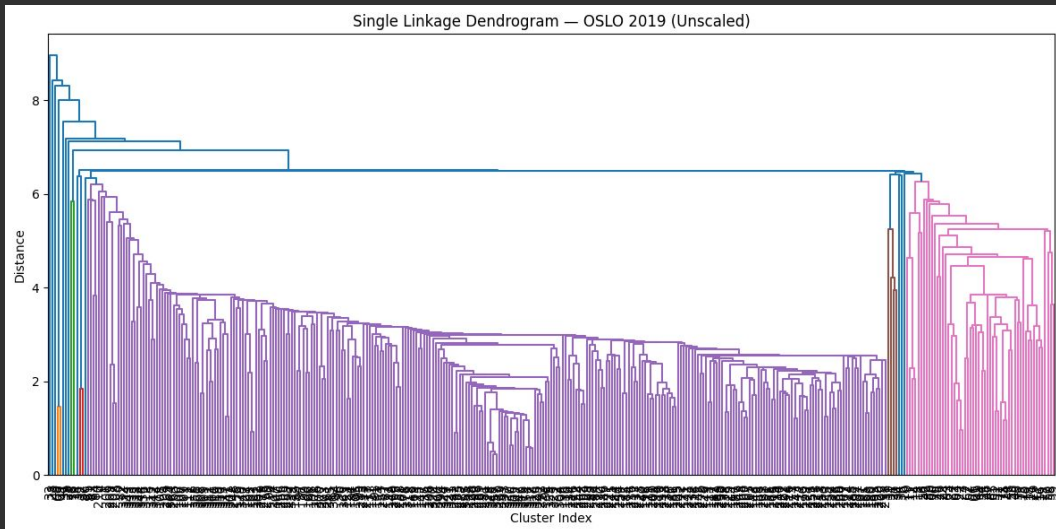




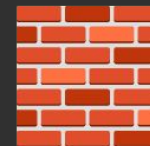
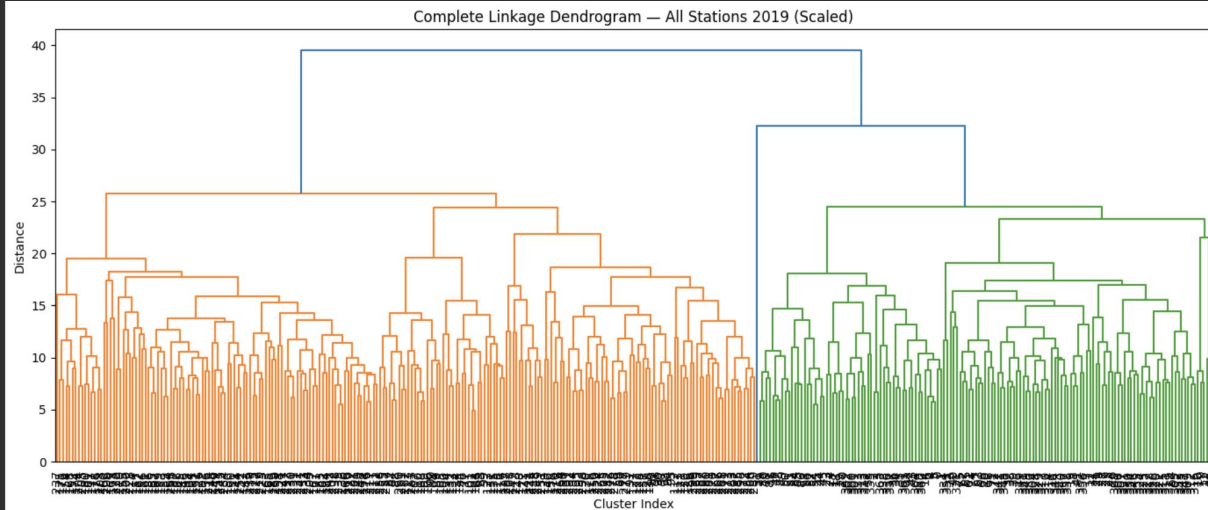
Single Linkage

Chain Effect/Elongated Clusters: Creates long, chain-like clusters that can stretch across the data space - you can see this clearly in both dendrograms where clusters form extended chains rather than compact groups

The Oslo 2019 example particularly shows this chaining effect with the long horizontal lines at low distances, while the "All Stations" plot demonstrates how single linkage can create very uneven cluster sizes with most points grouped into extended chains.



* analysis assistance from Claude AI, Perplexity AI, and ChatGPT via back-and-forth exchanges



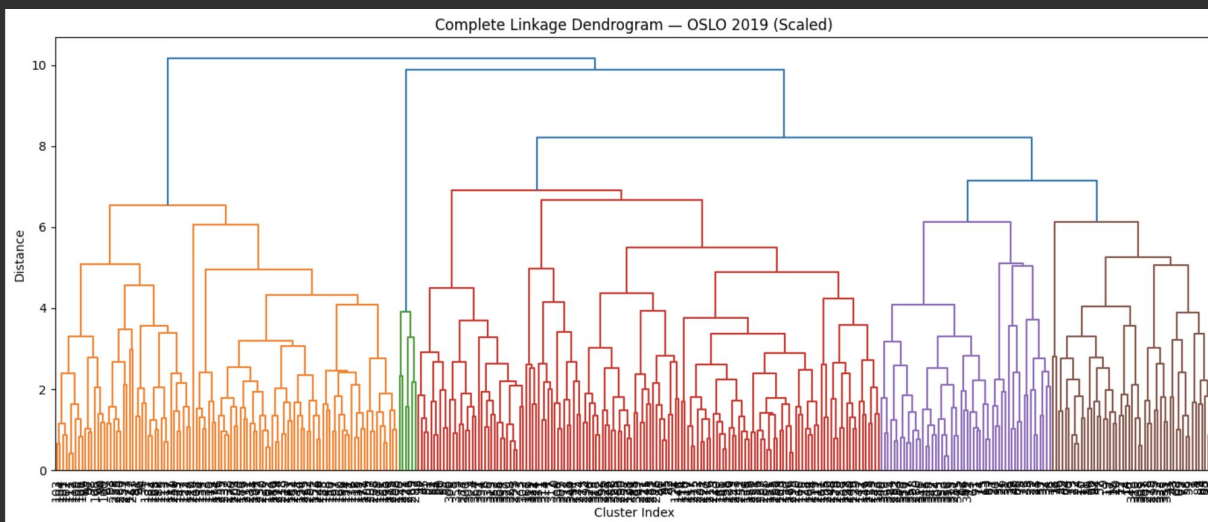
Complete Linkage

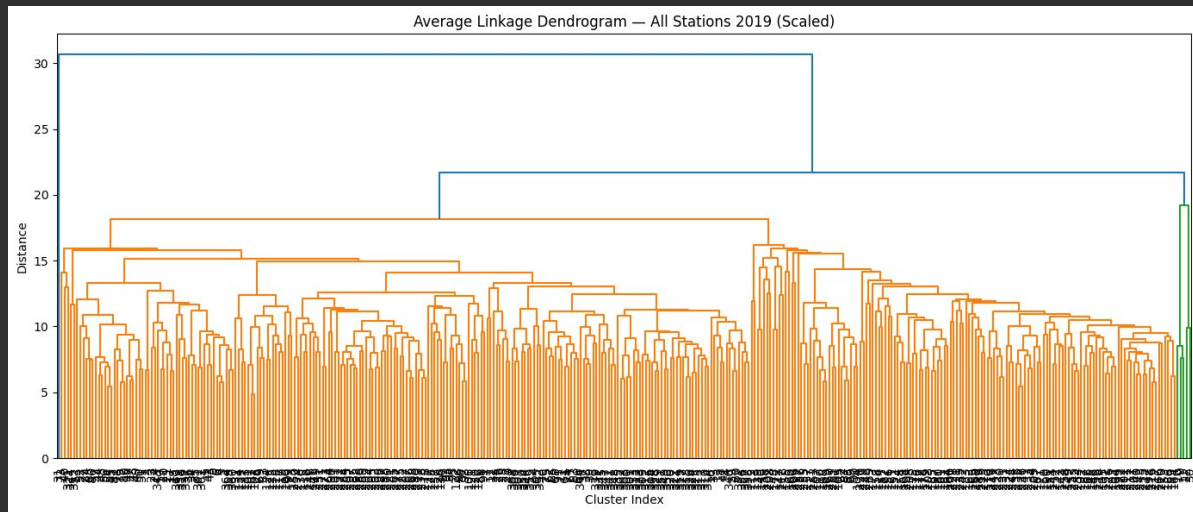
Compact, Spherical Clusters: Creates tight, well-separated clusters with similar sizes; the colored clusters in both dendrograms are much more balanced and compact compared to single linkage's chains

Higher Merge Heights: Clusters merge at higher distances because complete linkage waits until even the furthest points between clusters are reasonably close: thus, a taller, more spread-out vertical structure

Outlier Resistance: Less sensitive to outliers and noise because it considers the worst-case (maximum) distance when deciding merges, preventing premature cluster joining that single linkage suffers from

Compared with linear linkage, we see clearly defined, evenly-sized cluster groups (orange, green, blue, purple sections) with a much more organized, tree-like dendrogram structure. The Oslo example particularly shows how complete linkage creates distinct, well-separated clusters rather than sprawling chains.





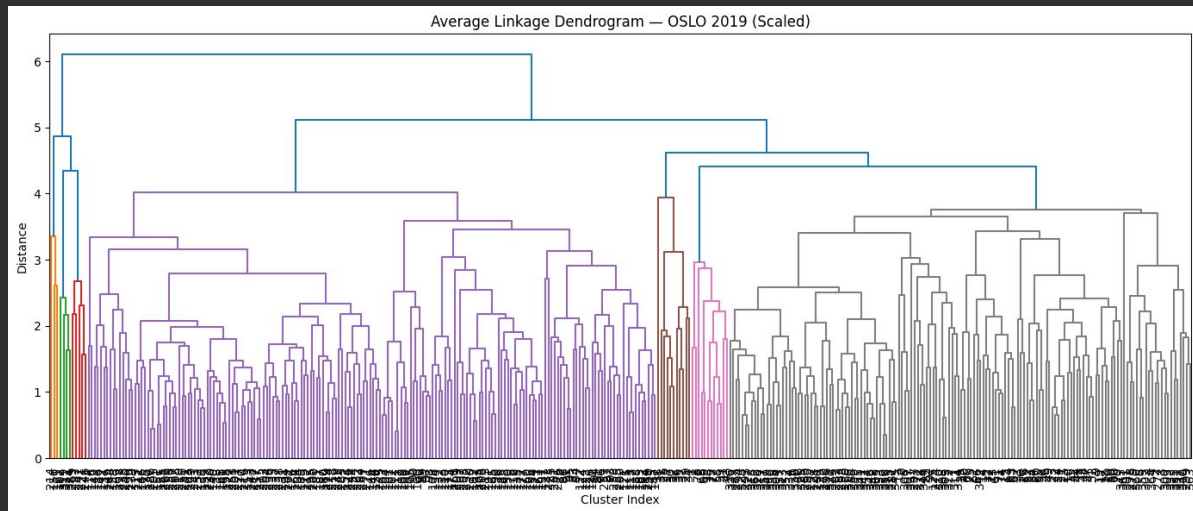
Average Linkage

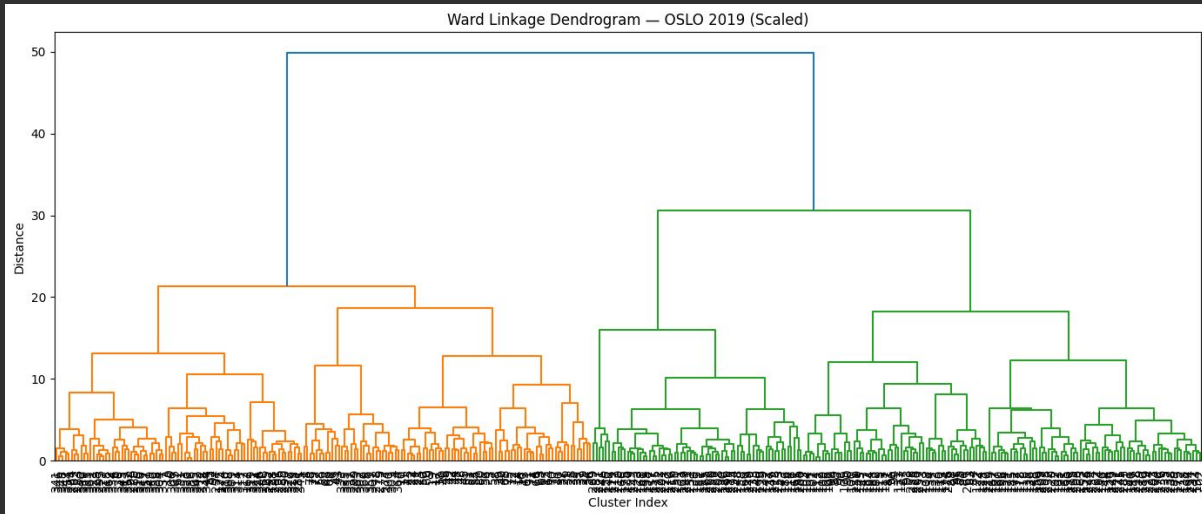
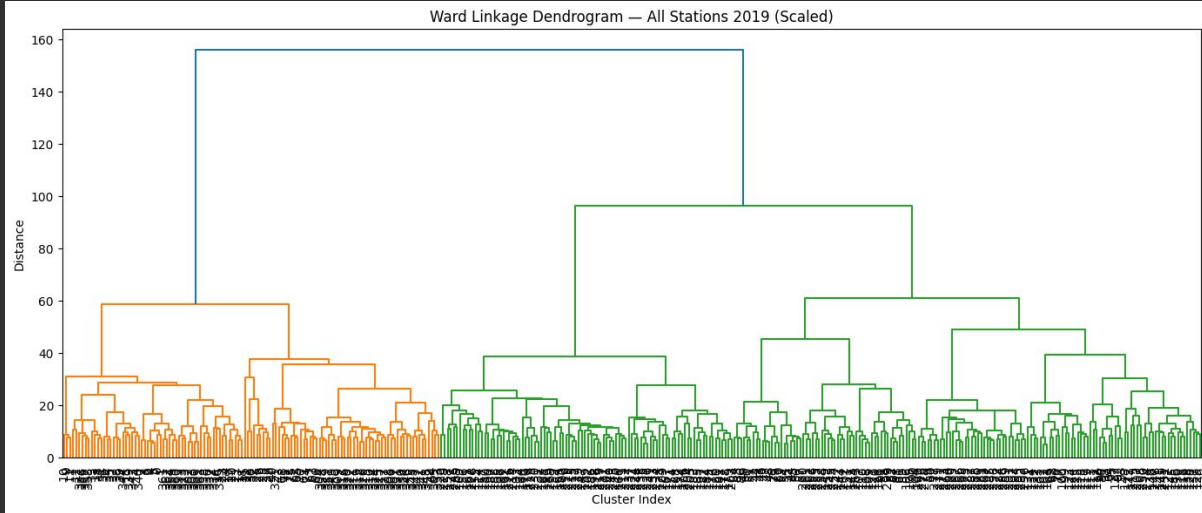
Balanced Compromise: Produces clusters that are neither as elongated as single linkage nor as compact as complete linkage - you can see moderately sized, reasonably well-separated clusters

Moderate Tree Balance: Creates dendrograms that are more balanced than single linkage but less perfectly symmetric than complete linkage - notice the mixed structure with some asymmetry but overall reasonable balance

Gradual Merge Heights: Shows a more gradual progression of merge distances compared to the extreme early merging of single linkage or the high-distance merging of complete linkage - evident in the intermediate vertical spacing

The Oslo example particularly demonstrates this "middle path" approach - you can see distinct cluster groups (purple, pink, gray sections) that are neither chained together like single linkage nor as tightly constrained as complete linkage.





Ward Linkage

Equal-Sized, Spherical Clusters: Strongly biased toward creating clusters of similar sizes.. Notice how evenly balanced the orange and green clusters are in both dendrograms; Because of *variance minimization focus*, ward linkage creates the most internally homogeneous clusters - each cluster has points that are very similar to each other

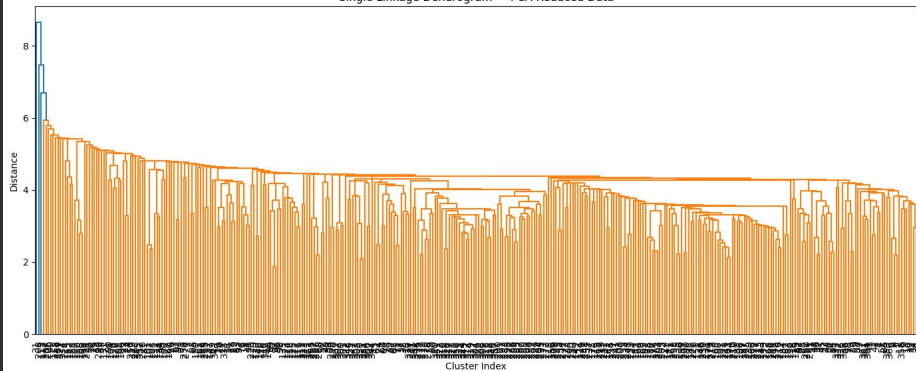
Dramatic Height Differences: Shows the most extreme vertical jumps in merge heights– indicating very distinct cluster separations and clear "stopping points" for cluster selection. The large height jumps make it easy to determine the optimal number of clusters by looking for these dramatic increases

The Ward dendrograms show the most dramatic and interpretable structure of all four methods. The huge vertical jumps are Ward's signature feature - they clearly indicate natural cluster boundaries. This makes Ward linkage particularly popular for exploratory data analysis because it tells you exactly where to "cut" the dendrogram to get meaningful clusters.



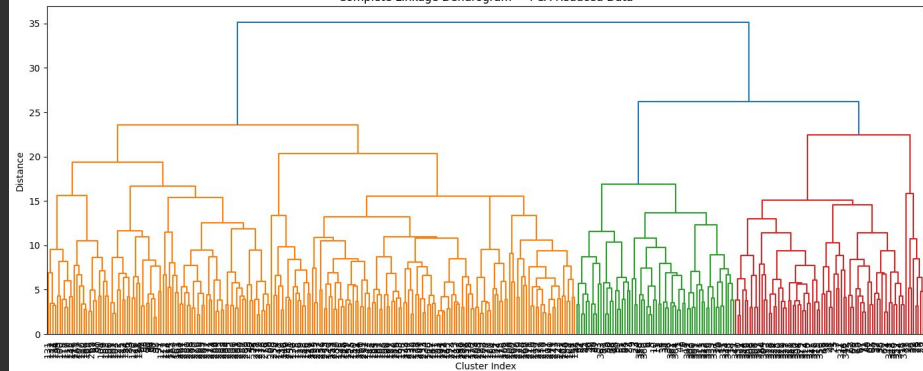
Single Linkage

Single Linkage Dendrogram — PCA-Reduced Data



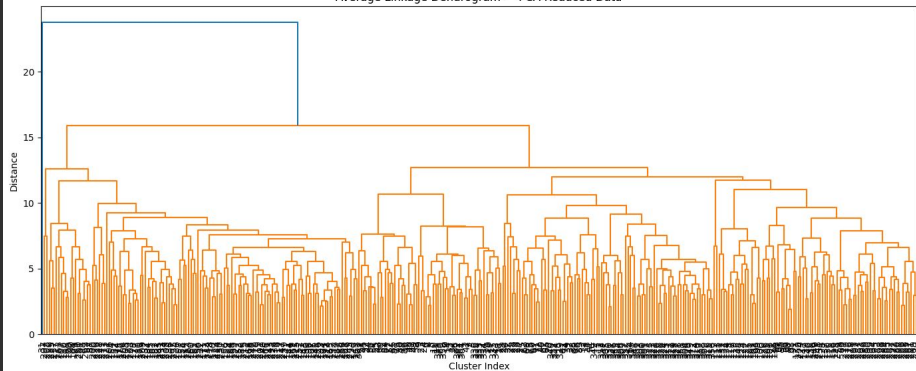
Complete Linkage

Complete Linkage Dendrogram — PCA-Reduced Data



Average Linkage

Average Linkage Dendrogram — PCA-Reduced Data

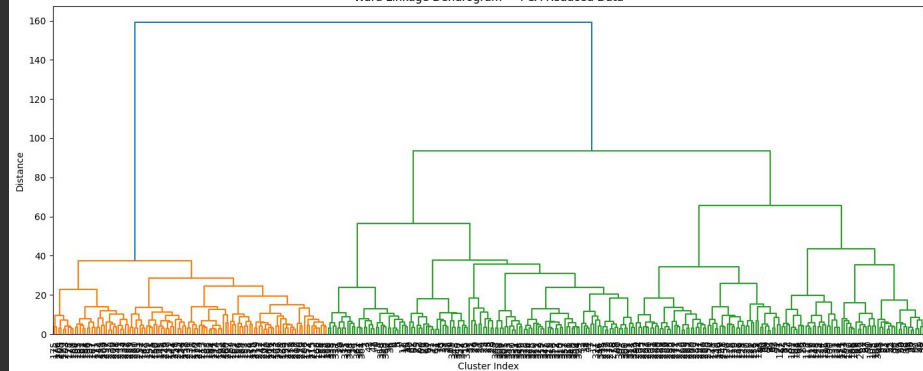


PCA: 9 components



Ward Linkage

Ward Linkage Dendrogram — PCA-Reduced Data



PCA Reduced Set (9 Components)

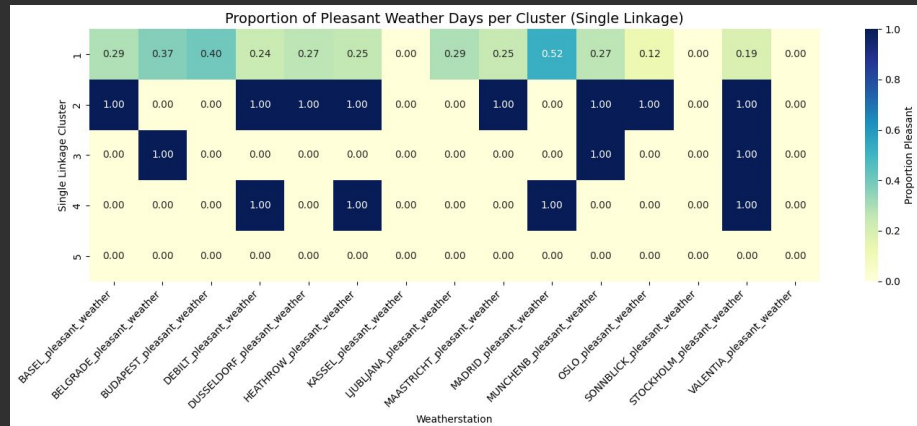
Running hierarchical clustering on the PCA-reduced dataset for all stations (2019) resulted in similar cluster shapes as before, but with simplified color groupings.

-  **Single linkage** still produced elongated, chain-like clusters, but now there is only a single outlier (blue) group and the rest are merged into one large cluster (orange).
-  **Complete linkage** resulted in three well-balanced groups that merge at higher distances, showing clearer separation.
-  **Average linkage**, similar to single, showed one distinct outlier group (blue) and a large merged cluster (orange).
-  **Ward linkage** behaved much like before, with two clearly separated clusters that merge only at a large vertical distance, indicating strong dissimilarity.

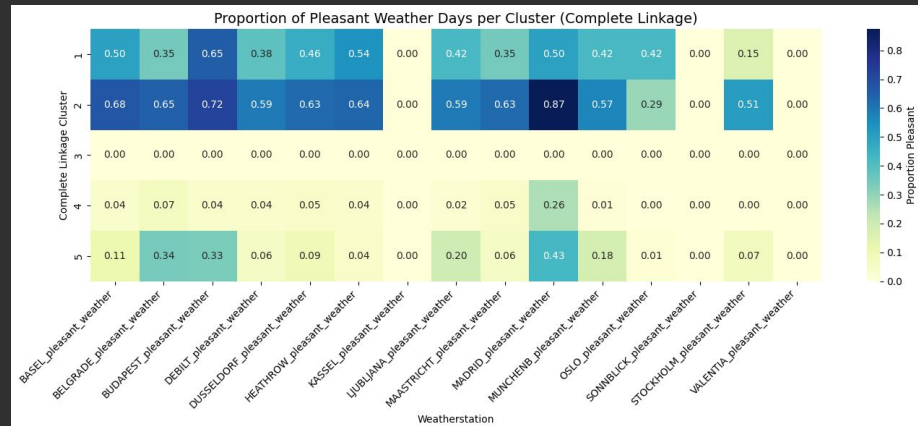
Interpretation: Applying PCA preserved the general structure of the data while reducing dimensionality and simplifying noise. This made the clustering results more interpretable, with cleaner group distinctions and fewer small, noisy clusters. ***The separation maintained by Ward and complete linkage suggests that the most important variance in the data (captured by PCA) still reflects meaningful groupings, while single and average linkage remain more sensitive to chaining effects and local similarities.*** PCA thus appears to enhance interpretability without distorting the overall relationships between observations.



Single Linkage



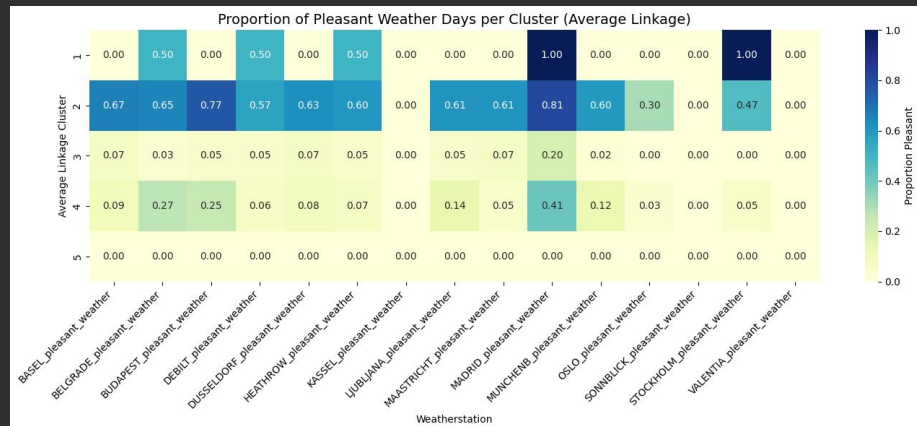
Complete Linkage



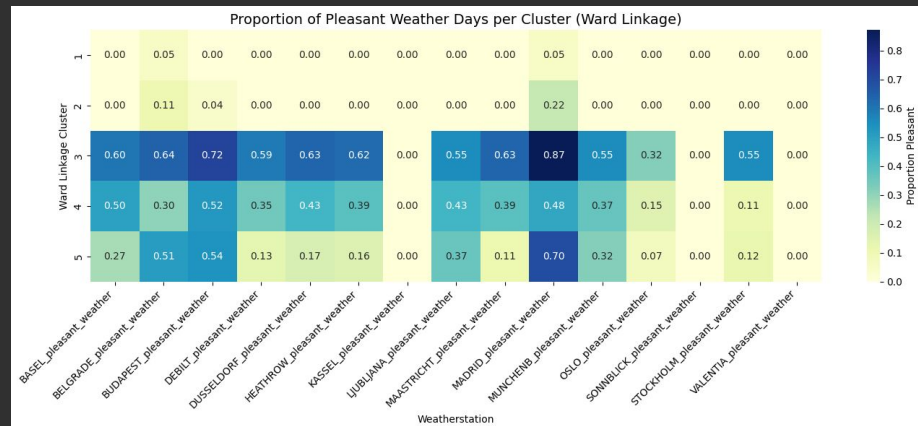
'pleasant_weather' distribution



Average Linkage



Ward Linkage



'Pleasant_weather' distributions by hierarchical clustering

Pleasant Weather Interpretation:

Analyzing the cluster-wise average of pleasant weather conditions revealed some key distinctions across linkage methods:

-  **Single linkage** produced clusters with the most *extreme* captures — several cities had 100% pleasant weather in one cluster and 0% in others, indicating high sensitivity to localized similarity (but possibly overfitting to specific weather profiles).
-  **Ward linkage**, in contrast, distributed high pleasant weather scores more evenly across clusters. While no single cluster captured all the “pleasant” days for a given city, the separation was more balanced, suggesting it grouped observations based on broader weather patterns rather than extremes.
-  **Average linkage** had a more mixed distribution — it included clusters with full captures for certain cities, like single linkage, but also had clusters where cities had moderate shares, indicating it may strike a middle ground between localized chaining (like single) and global structure (like Ward).

Interpretation:

These patterns suggest that single linkage may be overly sensitive to local outliers, while Ward linkage captures more general atmospheric patterns across stations. Depending on the goal (e.g., identifying *all* pleasant days vs. finding shared weather regimes), the choice of linkage and how clusters are interpreted may differ.